

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

LEARNING DIRECTED GRAPHICAL MODELS

SANJOY DASGUPTA, PROFESSOR

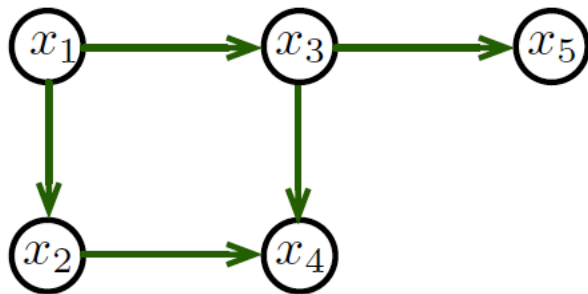
UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE



Fixed Structure, Fully Observable

Maximum-likelihood parameters can be found efficiently.



Fixed Structure, with Hidden Variables

Maximum-likelihood parameter estimation is typically NP-hard.

Example: mixtures of Gaussians.

Usual approach: EM algorithm.

Unknown Structure

Finding the maximum-likelihood structure is NP-hard.

Well, there is one special case: *branchings* (trees in which each node has ≤ 1 parent), often called *Chow-Liu trees*.

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- Can capture any probability distribution
- Intuitive factored form
- Popular for representing causal dependencies
- A pictorial language in which to specify families of probability models

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- Sampling: easy
- Computing likelihood/density: easy if fully-observable
- Inference: hard, but approximate methods are available
- Learning: many cases

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Suitability for (1) density estimation (2) generation (3) representation learning?